

Test 1

November 5, 2023

Problem 1

Restate the following in statements in an "if α then β " form so that α implies β .

- a) I will only mow the lawn if you pay me twenty dollars.
- b) Everyone who passed the test got an A in the class.
- c) Only the wise can understand this statement.

Problem 2

The only way for Dorothy to get to Oz is to follow the yellow brick road.

- a) What can you conclude if Dorothy got to Oz?
- b) What can you conclude if Dorothy follows the yellow brick road?

Problem 3

If P and Q are points in the interior of $\angle O$, prove that the segment PQ is in the interior of $\angle O$.

Problem 4

Prove that every line has a point on either side of it.

Problem 5

In $\angle AOB$, where $\angle AOB$ is NOT a straight angle, point P lies between A and B . Prove that P is in the interior of the angle.

Problem 6

Prove that if $AB \not\cong CD$ then $AB \neq CD$.

Problem 7

Prove that $0.\overline{9} = 1$.

Extra Credit

A line l does not contain any of the vertices of a triangle $\triangle ABC$. The line l intersects side AB , but does NOT intersect side BC . Prove that l must intersect the side AC .